

Wai & Watts — Portfolio Summary

New Zealand environmental data platform demonstrating Java/Spring Boot backend engineering, deterministic ingestion, dataset lineage, and grounded LLM explanation workflows.

Overview

Wai & Watts is a lineage-first environmental data platform designed to ingest public environmental and energy datasets, preserve explicit dataset lineage, and generate grounded natural-language explanations using a fact-pack architecture that prevents hallucination.

The project was intentionally built as a portfolio artifact to demonstrate production-grade backend engineering, ingestion pipeline design, and disciplined AI integration patterns.

Key Architectural Capabilities

Deterministic ingestion pipeline: Publisher spreadsheets are transformed into canonical contract CSV files before ingestion, ensuring reproducibility and protecting against publisher format drift.

Explicit dataset lineage model: Each ingestion produces an immutable dataset_release record linked to its originating dataset_source, allowing complete reconstruction of system state and safe re-ingestion.

Contract-driven architecture: All ingestion is based on normalized contracts rather than raw publisher artifacts, isolating ingestion logic from external variability.

Grounded LLM explanation architecture: LLMs operate exclusively on structured Fact Packs generated by deterministic database queries, preventing hallucination and ensuring explanations remain verifiable.

Full-stack implementation: Backend: Java, Spring Boot, Postgres. Frontend: React, TypeScript. Infrastructure: Docker, CI pipelines, CLI ingestion tooling.

Engineering Practices Demonstrated

- ingestion lifecycle modeling
- idempotent system architecture
- contract-based data ingestion
- operational reproducibility
- lineage-first database design
- safe AI integration boundaries
- production-style repository governance
- full local reproducibility

Why This Matters

Most portfolio projects demonstrate application development. Wai & Watts demonstrates system design discipline - including data lineage, ingestion correctness, architectural guardrails, and safe integration of AI into production-style systems.

These patterns are directly applicable to real-world data platform and backend engineering roles.

Status

Completed as a portfolio project. The system supports deterministic ingestion, lineage tracking, grounded explanation generation, and full local reproducibility.